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# Lagged Determinants of U.S. Output Growth

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## Abstract

This paper presents an autoregressive model of Real GDP with exogenous regressors that are connected to output in the Keynesian framework. A relatively standard/conventional statistical test suite is augmented with newly discovered modifications to the Kolmogorov–Smirnov test to facilitate time-series data and the interdependence brought by it. Building on the original validation methods that are rigorously tested over the decades, the paper manages to present moderate evidence that the estimator specified is efficient, well conditioned, and interpretable.

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# 1 Introduction

## 1.1 Overview

Real GDP is a prominent measure of output due to its simplicity, interpretability, and effectiveness in communicating the overall activity of an economy. Alongside academic settings, Real GDP has retained its relevance in politics as one of the go-to measures for public communications regarding economic growth and prosperity. As per its wide-spread adoption, the measure has accumulated a plethora of literature referencing it from a multitude of perspectives.

This paper explores the link between GDP growth and theoretically grounded interactions of GDP deficit, unemployment gap, and output gap, alongside other exogenous regressors. Ultimately, the paper aims to show that empirical observations of Real GDP growth in the U.S. economy can be modelled optimally with a **Gauss–Markov** compliant **Distributed Lag Model** (DLM) specification where the exogenous contributors are theoretical descriptors of output.

## 1.2 Hypothesis

In the history of economic equilibria, multiple models have proposed mathematically defined relationships between Real GDP and many indicators. Moreover, some interactions between said indicators have been formalized into economic models. This paper focuses on testing the predictive and explanatory power of two such indicators involving unemployment gap, output gap, and GDP deflator ( $P_t^{GDP}$ ):

- Unemployment Gap and Output Gap (**Okun-like**):  $O_t = (u_t - u_t^*) \times (Y_t - Y_t^*)$
- Unemployment Gap and GDP Deflator (**Phillips-like**):  $\Pi_t = (u_t - u_t^*) \times \Delta \log(P_t^{GDP})$

Alongside other exogenous regressors, this paper hypothesizes that a model of the form:

$$\Delta \log(Y_t) = C + \sum_{i=1}^n \beta_i X_{[i, t-1]} + \varepsilon_t \quad (1)$$

where  $X_{[i, t-1]}$  includes the above defined interactions has a **BLUE** OLS estimator with nonzero interaction coefficients.

## 1.3 Theoretical Background

We consider two equations to base our selection of interactions. First, the **Phillips Curve** describes the relationship between inflation and unemployment. It can be expressed in terms of expected inflation and the unemployment gap as follows:

$$\pi_t = \mathbb{E}_t[\pi_{t+1}] - \alpha(u_t - u_t^*) + \varepsilon_t \quad (2)$$

where  $\pi_t$  is the inflation rate at time  $t$ ,  $\mathbb{E}_t[\pi_{t+1}]$  is the expected inflation rate at time  $t + 1$ ,  $\bar{u}_t$  is the unemployment gap,  $\alpha$  is a positive constant, and  $\varepsilon_t$  is a random error term. Equation 2 displays a negative relationship between inflation and unemployment gap with a factor of  $\alpha$ . The inflation rate  $\pi_t$  in logarithmic space can be approximated by  $\log(1 + \pi_t) = \Delta \log(P_t)$  for sufficiently small  $\pi_t$ . (up to 8-10%) The presented interaction substitutes the deflator for price levels and uses  $\Delta \log(P_t^{GDP})$  as a deflator-based inflation term. In principle, we expect (and hypothesize) that the pseudo-inflation term defined will again have a negative relationship with unemployment gap with a non-zero factor.

We can use the Phillips Curve to connect inflation and unemployment; however, the connection of this interaction to our regression target is not yet established. Utilizing the **Fischer Equation**, we can relate inflation to nominal interest rates, and express output as a function of inflation using the **IS Curve**.

$$\begin{aligned} r_t &= i_t - \pi_t \\ \Rightarrow Y_t &= Y^* - \alpha(r_t - r_t^*) \end{aligned} \tag{3}$$

Having established  $Y_t$  as a function of  $\pi_t$ , we can expect to see a relationship in growth rates as well.

Next, we consider **Okun's Law**, which describes the relationship between unemployment and output (Real GDP):

$$\frac{(Y_t - Y_t^*)}{Y_t^*} = -\gamma(u_t - u_t^*) \tag{4}$$

where  $Y_t$  is the actual output,  $Y_t^*$  is the potential output,  $\gamma$  is a positive constant, and  $(u_t - u_t^*)$  is the unemployment gap. Okun's Law can be algebraically rearranged to express  $Y_t$  in levels:

$$\begin{aligned} \frac{Y_t}{Y_t^*} - 1 &= -\gamma(u_t - u_t^*) \\ Y_t &= Y_t^*(1 - \gamma(u_t - u_t^*)) \end{aligned} \tag{5}$$

The above transformation connects output to unemployment and establishes a negative relationship. A simple yet fundamental observation is that non-zero relationships of  $P_t$ , and  $(u_t - u_t^*)$  are clearly defined in theory. Interactions defined with these variables are therefore potential candidates to explain the output growth. Therefore, an empirical relationship can be expected; and such relationship can be analyzed to determine the differences between theoretical and observed interactions.

## 1.4 Literature Review

The literature review in this paper is aimed to justify two main choices made for our methodology and selection of variables:

- Selection of the exogenous variable set
- Usage of an *ARX* model

The theoretical relationships between exogenous variables and output were displayed in Section 1.3. In addition to said demonstration, there are numerous papers investigating these relationships. Primarily, “The Relationship between GDP and Unemployment Rate in the U.S.” (Mandel and Liebens, 2019) employs a very similarly structured model to conduct correlation analysis. The paper demonstrates the existence of significant correlations  $\rho_{Y,CPI}$  and  $\rho_{Y,u}$  (among other variables) through multiple regression analysis. However, an important note regarding their application is the usage of  $CPI_{t-0}$  and  $u_{t-2}$  which do not align the orders selected in our paper.<sup>1</sup>

Another relevant paper, “GDP Forecasting: Machine Learning, Linear or Autoregression?” (Maccarrone et al., 2021), compares the performance of various simple regression models and machine learning algorithms in forecasting GDP. The authors conclude that *ARX* in their specification is the best model in “one-step-ahead” and comparable to the best in “multi-step-ahead” forecasts. This paper confirms the intuitively autocorrelated nature of GDP, and the fact that exogenous variables have the ability to improve fit quality.

Although there are many other papers investigating GDP modelling from simple regression to deep learning, the two papers mentioned above capture the reasoning behind the methodology design and model selection in our paper.

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<sup>1</sup>Refer to Section ?? for the order selection of this paper.

## 2 Data

The initial model specification includes a wide range of macroeconomic indicators alongside the interactions. Observations for all series span from 1989Q4 to 2025Q1 and the frequency is quarterly. All series are sourced from the St. Louis Federal Reserve Economic Data (FRED) database.

| Variable                                    | Units                                       | FRED ID                         |
|---------------------------------------------|---------------------------------------------|---------------------------------|
| Real GDP ( $Y$ )                            | Billions of Chained 2017 Dollars (Adjusted) | <a href="#">GDPC1</a>           |
| Personal Consumption Expenditures ( $pce$ ) | Billions of Chained 2017 Dollars (Adjusted) | <a href="#">PCECC96</a>         |
| Private Domestic Investment ( $pdi$ )       | Billions of Chained 2017 Dollars (Adjusted) | <a href="#">GPDIC1</a>          |
| Government Expenditures ( $govexp$ )        | Billions of Chained 2017 Dollars (Adjusted) | <a href="#">GCEC1</a>           |
| Output Deflator ( $odef$ )                  | Chained Index 2017=100 (Adjusted)           | <a href="#">GDPDEF</a>          |
| Natural Unemployment Rate ( $u^*$ )         | % (Not Adjusted)                            | <a href="#">NROU</a>            |
| Unemployment Rate ( $u$ )                   | % (Adjusted)                                | <a href="#">LRUN64TTUSQ156S</a> |
| Potential Output ( $Y^*$ )                  | Billions of Chained 2017 Dollars (Adjusted) | <a href="#">GDPPOT</a>          |

Table 1: Macroeconomic series used in the model along with their FRED IDs.

Using the series in Table 1, we construct our target variable and regressors as follows:

- Target Variable:  $\Delta \log(Y_t)$
- Regressors:

$$X_{[t, 1]} (pce): \Delta \log(pce_{t-1})$$

$$X_{[t, 2]} (pdi): \Delta \log(pdi_{t-1})$$

$$X_{[t, 3]} (govexp): \Delta \log(govexp_{t-1})$$

$$X_{[t, 4]} (ugap): u_{t-1} - u_{t-1}^*$$

$$X_{[t, 5]} (ogap): \log(Y_{t-1}) - \log(Y_{t-1}^*)$$

$$X_{[t, 6]} (okun): ugap \times ogap$$

$$X_{[t, 7]} (phillips): ugap \times \Delta \log(odef_{t-1})$$

All variables that are observed in levels are transformed into log differences while both the gap variables and interactions are not differenced. In our specification, log differencing is used to clean the regressors of slow-moving trends and potential multicollinearity; which is mostly an artifact of data in pure levels. The gaps and interactions are already detrended to an extent by their construction.

## Distributions of Regressors

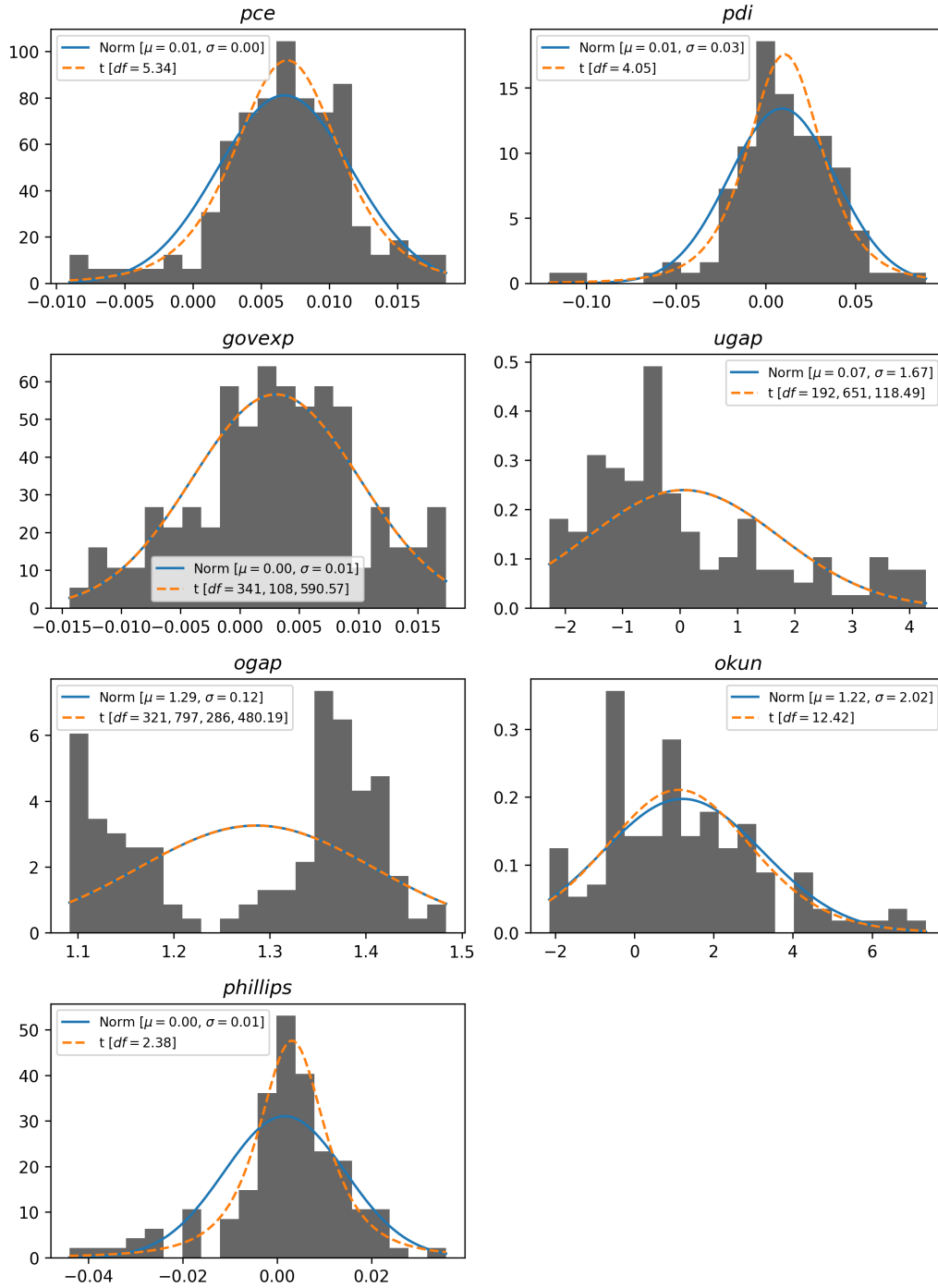


Figure 1: Density histograms of regressors.

### 3 Methodology

The proposed methodology of this paper can be split into four main components:

- Data Preparation and Examination
- Model Estimation
- BLUE Evaluation
- Post-Fit Validation

Each component is detailed in the following subsections.

#### 3.1 Data Preparation and Examination

The preparation step consists of using the series defined in Table 1 to create regressors and target as listed. Vectorized transformations are applied to the raw data to create the necessary variables. The differencing process requires each observation to have a predecessor; therefore the first observation is lost in the transformation. This pushes our observation window from 1989Q4–2025Q1 to 1990Q1–2025Q1.

After the variables are constructed, we shift all regressors by one period to produce the final dataset. This ensures that all regressors at time  $t$  are known; making the model have predictive validity rather than being a purely explanatory specification. Having shifted the regressors, we again lose the first observation in the dataset, pulling the final observation window to 1990Q2–2025Q1 with a total of 140 observations.

##### 3.1.1 Design Matrix Formation

$X$  is formed by converting the model equation specified in Equation 1 into matrix form. The inner product operation of OLS estimation can be written as a matrix multiplication of observations and coefficients:

$$\underbrace{\begin{bmatrix} 1 & X_{1,2} & \cdots & X_{1,m+1} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & X_{n,2} & \cdots & X_{n,m+1} \end{bmatrix}}_{\text{Design Matrix}} \times \begin{bmatrix} C \\ \beta_1 \\ \vdots \\ \beta_{m+1} \end{bmatrix} \quad (6)$$

Where  $n$  is the number of observations and  $m$  is the number of features (including the intercept). Using our observation count  $n = 140$  and number of regressors  $m = 7$ , the design matrix  $X$  takes the dimensions of  $140 \times 8$ .



### 3.1.2 Regressor Examination

While the model order will not be adjusted based on the results, the Cross Correlation Function (CCF) of each regressor with the target variable will be examined to confirm the appropriateness of the lag structure. The **Box–Jenkins Prewhitening** procedure will be applied to expose a clearer signal of the underlying cross-correlations. Box–Jenkins Prewhitening transforms the exogenous variables into residuals of their  $AR$  representation, then uses the  $\beta$  coefficient to re-scale the dependent variable. (Hackhard and Romer, 2025) Formally, the  $AR(1)$  variant of the process can be shown as:

$$\begin{aligned}\hat{X}_i &= (L^1 X_i)\beta_i \\ \hat{y}_i &= y\beta_i \\ \rho_{y,i} &= \text{corr}(\hat{y}_i, X_i - \hat{X}_i)\end{aligned}\tag{7}$$

Where  $L^1$  is the lag-1 operator.

Additionally, the **Variance Inflation Factor** (VIF) of all regressors will be checked to ensure multicollinearity is not present. VIF is computed as:

$$VIF_{X_i} = \frac{1}{1 - R_{X_i}^2}\tag{8}$$

Where  $R_{X_i}^2$  is the coefficient of determination of the regression of  $X_i$  on all other columns of  $X$ . The common threshold for multicollinearity,  $VIF > 10$ , will be used.

## 3.2 BLUE Evaluation

After estimating the model coefficients, we evaluate the Gauss–Markov conditions to confirm whether the resulting fit is BLUE. Previous stages of the methodology involve checks for the pre-fit Gauss–Markov conditions:

- **Linearity of Parameters:** by definition the model employed is a linear combination of regressors and coefficients. Therefore, this condition is satisfied at the stage of model selection.
- **Full Rank:**  $\text{rank}(X) = k$  must hold for  $X^\top X$  to be invertible. Therefore, this condition must hold for the model to be estimable via OLS.

The remaining Gauss–Markov conditions concern  $\varepsilon$  and are verified post-estimation:

- **Spherical Errors:** This condition holds if  $\text{Var}(\varepsilon) = \sigma^2 I_n$ ; implies that the residuals are homoskedastic and uncorrelated. The two criteria can be tested separately using the **Breusch–Pagan Test** for homoskedasticity and the **Breusch–Godfrey Test** for autocorrelation.

- **Strict Exogeneity:** In textbook terms strict exogeneity requires  $\mathbb{E}[\varepsilon_i | X] = 0$ . However, this condition is not directly testable. It’s often assumed to hold given weak exogeneity holds, and the design matrix  $X$  shows no direct signs of endogenous interactions. A **RESET Test** will be employed to check for model misspecification and omitted nonlinearities. Moreover, the **Harvey–Collier Test** will be used to further assess model specification. Strong exogeneity will then be assumed, and therefore any finding of BLUE will be referred to as “approximate” under the modelling assumptions.

### 3.3 Post-Fit Validation

After estimating the model and evaluating the Gauss–Markov conditions, regression coefficients and residuals will be evaluated. Alongside understanding variable significances and outliers, this setep will potentially highlight the sources of issues that may arise with the BLUE evaluation.

#### 3.3.1 Coefficient Significance

Coefficients will be tested for significance both individually and jointly. Individual significance will be tested with *t*-tests, while model-wise joint significance will be tested with an *F*-test. Additionally, the joint significance of coefficient subsets will be tested using **Wald Tests**.

#### 3.3.2 Residual Analysis

The residuals of a model are very important both in predictive terms, and as tools of analysing the model fit. Therefore, multiple aspects of the residuals will be examined before deeming the model valid/efficient.

To begin with, the fitted values will be plotted against the observed target to display the directionality and scale of residuals in a visual manner. This will be complemented by a QQ plot and a **Shapiro–Wilk Test** to assess the normality of residuals. Normality is not a Gauss–Markov condition, but it is important for inference.

Finally, potential outliers will be examined using **Leverage** and **Cook’s Distance**. Given the nature of our target, we can immediately expect that the 2008 Financial Crisis and 2020 COVID-19 shocks will be visible in this portion of the analysis. If found to be detrimental to the model generalization, removal of outliers will be considered as required.

## 4 Results

### 4.1 Data Examination

With the design matrix constructed as per Section 3.1.1, we can compute the inter-regressor interactions with VIF and the regressor-target relations with CCF. Our initial VIF computations are as follows:

| Regressor       | VIF  |
|-----------------|------|
| <i>pce</i>      | 1.73 |
| <i>pdi</i>      | 1.61 |
| <i>govexp</i>   | 1.43 |
| <i>ugap</i>     | 1.45 |
| <i>ogap</i>     | 1.85 |
| <i>okun</i>     | 2.86 |
| <i>phillips</i> | 2.76 |

Table 2: Variance Inflation Factor (VIF) of all regressors.

All VIF values are well below the threshold; and we can safely conclude that multicollinearity is not a concern in our design matrix.

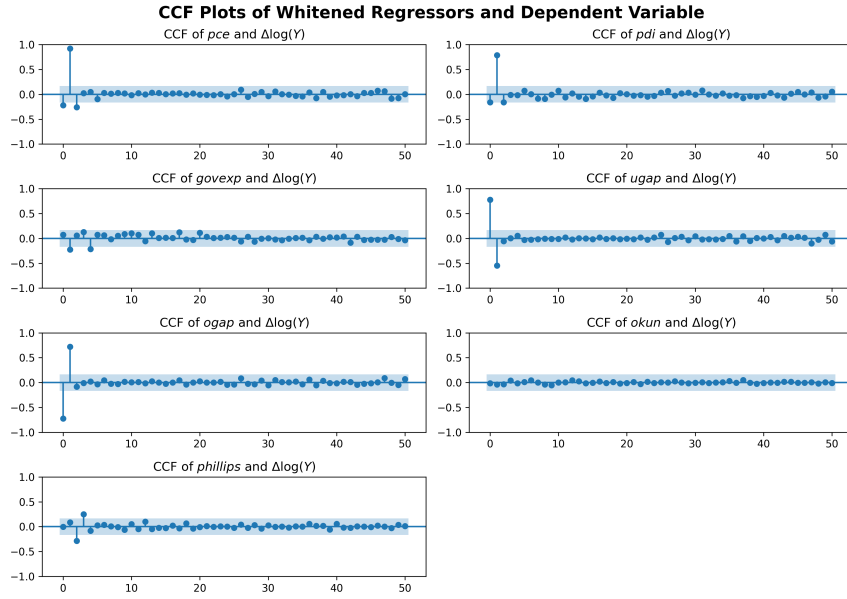


Figure 2: CCF plots of the whitened exogenous variables and the dependent variable.

The CCF results all but one regressor has at least one significant lagged correlation with the target. The regressors *pce*, and *pdi* are direct contributors to GDP. As expected, they show strong positive correlations at lag-1. Interestingly, we see that *ugap* has a significant positive correlation at lag-0; which is counterintuitive given the theoretical negative relation.

Moreover, we see that almost all variables with significant lags have a sign changed significant correlation immediately after the initial lag that's significant. This might be an indicator of cyclical behavior, or “response functions” with overshooting.

## 4.2 Model Estimation

Having confirmed the design matrix is sufficiently well-behaved, we proceed with the estimation of the initial model. The fit results in the coefficients:

| Variable        | Coefficient | t-test <i>p</i> |
|-----------------|-------------|-----------------|
| <i>C</i>        | -0.0024     | 0.8356          |
| <i>pce</i>      | -0.0733     | 0.4732          |
| <i>pdi</i>      | 0.0221      | 0.5193          |
| <i>govexp</i>   | 0.0903      | 0.5482          |
| <i>ugap</i>     | 0.0018      | 0.0044          |
| <i>ogap</i>     | 0.0058      | 0.5307          |
| <i>okun</i>     | 0.0011      | 0.1457          |
| <i>phillips</i> | -0.0139     | 0.1954          |

Table 3: Estimated Coefficients of the Initial Model.

Table 5 clearly shows that the model is not well-specified. With the only significant variable being *ugap*, we see that the model lacks predictive and explanatory power. The overall model fit statistics confirm this observation:

| Statistic               | Value                             |
|-------------------------|-----------------------------------|
| $R^2$                   | 0.0877                            |
| Adjusted $R^2$          | 0.0315                            |
| RMSE                    | 0.107                             |
| F-statistic             | 1.5612                            |
| p-value ( $F$ )         | 0.1426                            |
| $X^\top \hat{\epsilon}$ | $-8.79 \times 10^{-14} \approx 0$ |

Table 4: Overall model fit statistics.

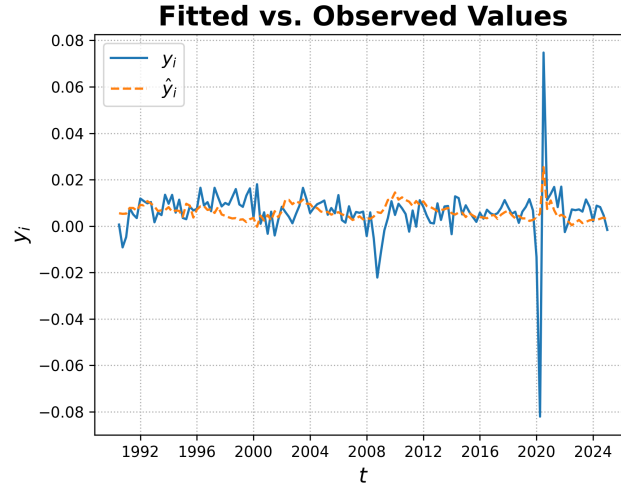


Figure 3: Fitted vs. Observed Values

With an  $R^2$  of only 8.77% and a failing  $F$ -test, we can conclude an adjustment of the model specification is necessary. For further confirmation, we ran a RESET test which resulted in a strong rejection with  $p = 3.8 \times 10^{-14}$ . However, another noticeable observation is the extreme outliers in Figure 8. As discussed in Section 3.1, COVID-19 proves to be a strong hurdle for the model while the 2008 is also a visible outlier to a lesser extent.

Through leverage and Cook's distance, we will confirm the influence of these outliers.

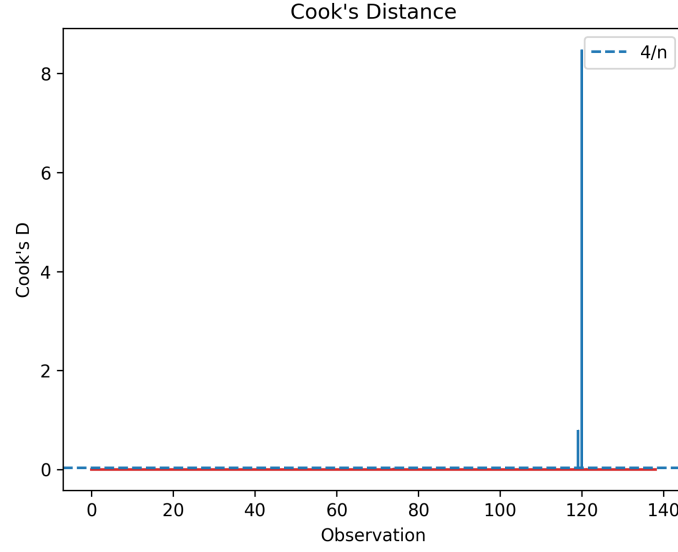


Figure 4: Cook's Distance of Initial Model Residuals.

We immediately see that the COVID-19 shocks dominate the influence statistics.  $D_{2020Q3} = 8.48$  and  $D_{2020Q2} = 0.801$  are the only visible points due to the significant shift in scale.

#### 4.2.1 Variable Distribution Equality

The standardized design matrix was used to conduct KS tests to confirm the equality of distributions. For demonstration purposes, both a simple KS test, and a bootstrapped test with 1,000 iterations were performed.

| KS Test Matrix ( $\alpha = 0.05$ ) |                                                      |                                                      |                                                      |
|------------------------------------|------------------------------------------------------|------------------------------------------------------|------------------------------------------------------|
| $y_{t-1}$                          | Fail to Reject $H_0$<br>$p = 1.0000$<br>$C = 0.0000$ | Fail to Reject $H_0$<br>$p = 0.8674$<br>$C = 0.0714$ | Reject $H_0$<br>$p = 0.0160$<br>$C = 0.1857$         |
| $CPI_{t-1}$                        | Fail to Reject $H_0$<br>$p = 0.8674$<br>$C = 0.0714$ | Fail to Reject $H_0$<br>$p = 1.0000$<br>$C = 0.0000$ | Fail to Reject $H_0$<br>$p = 0.0857$<br>$C = 0.1500$ |
| $u_{t-1}$                          | Reject $H_0$<br>$p = 0.0160$<br>$C = 0.1857$         | Fail to Reject $H_0$<br>$p = 0.0857$<br>$C = 0.1500$ | Fail to Reject $H_0$<br>$p = 1.0000$<br>$C = 0.0000$ |
|                                    | $y_{t-1}$                                            | $CPI_{t-1}$                                          | $u_{t-1}$                                            |

Figure 5: KS Test Matrix

| Bootstrapped KS Test Matrix ( $\alpha = 0.05$ ) |                                                      |                                                      |                                                      |
|-------------------------------------------------|------------------------------------------------------|------------------------------------------------------|------------------------------------------------------|
| $y_{t-1}$                                       | Fail to Reject $H_0$<br>$p = 1.0000$<br>$C = 0.0000$ | Fail to Reject $H_0$<br>$p = 0.9990$<br>$C = 0.0714$ | Fail to Reject $H_0$<br>$p = 0.1379$<br>$C = 0.1857$ |
| $CPI_{t-1}$                                     | Fail to Reject $H_0$<br>$p = 0.9970$<br>$C = 0.0714$ | Fail to Reject $H_0$<br>$p = 1.0000$<br>$C = 0.0000$ | Fail to Reject $H_0$<br>$p = 0.3926$<br>$C = 0.1500$ |
| $u_{t-1}$                                       | Fail to Reject $H_0$<br>$p = 0.1469$<br>$C = 0.1857$ | Fail to Reject $H_0$<br>$p = 0.4006$<br>$C = 0.1500$ | Fail to Reject $H_0$<br>$p = 1.0000$<br>$C = 0.0000$ |
|                                                 | $y_{t-1}$                                            | $CPI_{t-1}$                                          | $u_{t-1}$                                            |

Figure 6: Bootstrapped KS Test Matrix

We observe that the only difference in results between the two tests is in the  $\{y_{t-1}, u_{t-1}\}$  pair. Both tests use  $H_0: F_x = F_y$  therefore failing to reject means there was not enough evidence to refute the equality of distributions. However, it is important to note that these tests are not proof of equality, but rather an indication that the distributions are similar enough to not be statistically different. In this paper, we proceed respect the results of the bootstrapped test (Figure 6) and assume approximate equality of distributions for all variable pairs.

**NOTE:** The p-values annotated in the Standard KS matrix are approximations derived by a 101-term Taylor expansion. (Expansion is ill-defined at  $D = 0$  for an even number of terms) The test results are evaluated by a comparison of statistics and are not supposed to yield concrete p-values.

### 4.3 Model Fit

The fit was performed using Ordinary Least Squares (OLS) on the matrix expansion of Equation ???. The estimation yielded the following coefficients:

| Variable              | Coefficient |
|-----------------------|-------------|
| $C (\beta_0)$         | 16,257.27   |
| $y_{t-1} (\beta_1)$   | 3,795.25    |
| $CPI_{t-1} (\beta_2)$ | 14.83       |
| $u_{t-1} (\beta_3)$   | 65.54       |

Table 5: Estimated coefficients of the ARX(1, 1) model.

With a normalized design matrix, the intercept column expectedly has to take on the re-scaling of inputs. Also, the strong influence of  $y_{t-1}$  is inline with the autocorrelation analysis in Section ???. The only notable observation is the positive coefficient of  $u_{t-1}$ , which both by theory and correlation analysis should have been a negative value. This anomaly is likely attributable to the normalization step introducing negative values into an otherwise positive-only variable.

The overall model fit statistics are as follows:

| Statistic               | Value                            |
|-------------------------|----------------------------------|
| $R^2$                   | 0.9972                           |
| Adjusted $R^2$          | 0.9971                           |
| RMSE                    | 201.30                           |
| MAPE                    | 0.62%                            |
| F-statistic             | 12,002.50                        |
| p-value ( $F$ )         | $\approx 0$                      |
| $X^\top \hat{\epsilon}$ | $-6.25 \times 10^{-9} \approx 0$ |

Figure 7: Overall model fit statistics.

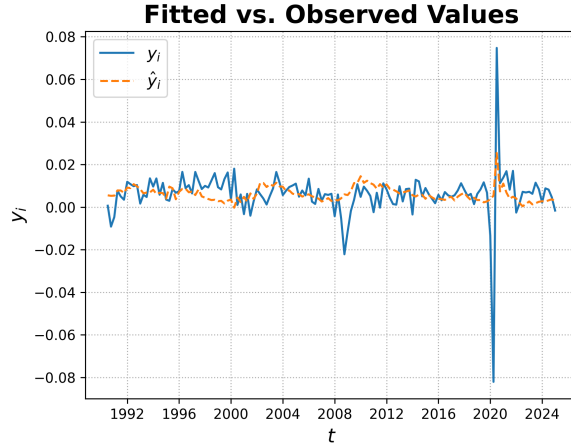


Figure 8: Fitted vs. Observed Values

All metrics indicate a good fit to in-sample data. However, the extremely high  $R^2$  and F-statistic are clear signs of overfitting. Overall, the fit behavior indicates a model where coefficients are interpretable, but predictions should be handled with care. Figure 8, displays 2 expected locations where residuals noticeably deviate from 0:

- The 2008 Financial Crisis
- The 2020 COVID-19 Shock

As expected from a generalized model, the shocks cause an increase in error. However, it's arguable that the model still responds to the shocks more than what would be expected from a well-generalized estimator. With the highly autocorrelated  $y$  and the presence of  $y_{t-1}$  in the design matrix, the unregularized  $ARX$  model expectedly "follows" the shocks. In fact, despite the reasonable fit statistics, the use of **Ridge** or **Lasso** regularization would yield better generalization for the tradeoff of a slightly reduced  $R^2$ .

To test the predictive performance of the model, a quick test with the only available out-of-sample data (Q2 2025) was conducted. A prediction of \$23,632.14 Bn was achieved against a true value of \$23,770.98 Bn. This results in a percentage error of  $-0.58\%$  which is within the expected range by MAPE.

#### 4.4 Residual Analysis

Given the fit information above, we can safely expect well-behaved residuals. However, a thorough analysis of residuals is of course necessary to confirm the validity of OLS assumptions. Plotting



residuals over time presents the opportunity to inspect the exact magnitudes of the errors in the shock periods discussed in Section 4.3.

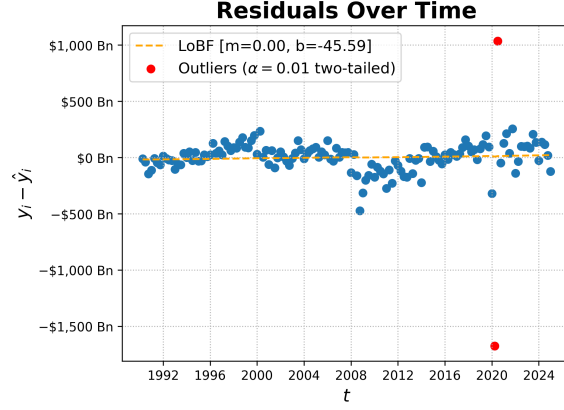


Figure 9: Residuals over time.

Figure 9 outlines two abnormalities in residuals using a 0.99 confidence interval to define outliers. Both extremes occur in the COVID-19 period, but looking at the 2008 period, we clearly see an overestimation of  $y$ . Inspecting the line of best fit shows us that the residuals are centered around 0, with a slope of 0. The residuals are heavily concentrated around 0, indicating a tighter empirical distribution compared to a Normal distribution. Fitting a Normal and t-student distribution to the residuals confirms this observation.

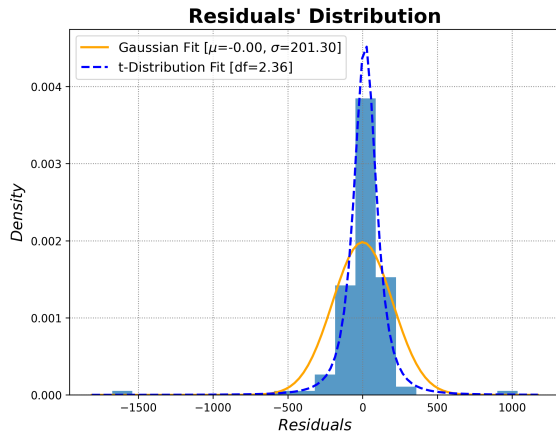


Figure 10: Residual density with fitted distributions.

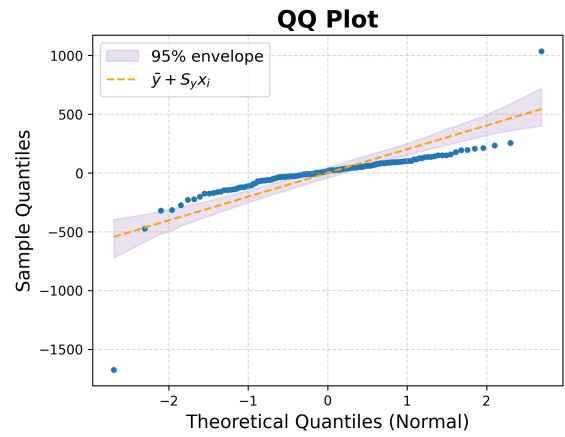


Figure 11: QQ Plot of Residuals vs.  $N(0, 1)$  quantiles.

The QQ plot of residuals against the Normal distribution (Figure 11) shows that most observations lie on a line with a noticeably smaller slope than the reference line  $\bar{y} + S_y x_i$ , indicating that the bulk of the residuals is less dispersed than a Normal distribution with the same mean and standard deviation. However, there are extreme points in both tails that lie well outside the 95% simulation envelope, reflecting occasional large shocks. Together with the fitted Student- $t$  distribution with  $\hat{\text{df}} \approx 2.36$ , this suggests a very concentrated core with heavier-than-Normal tails rather than Gaussian residuals. As an added confirmation, the Shapiro–Wilk test rejects normality with  $W = 0.62$  and  $p = 1.81 \times 10^{-17} \cong 0$ .

As a final descriptor of residual behavior, we inspect the stationarity using the ADF test. Knowing the residuals are zero-centered, we fit an ADF regression with no deterministic terms. With  $ADF = -2.25$  we obtain a  $p$ -value of 0.02 and reject the null hypothesis of a unit root at the 5% significance level. This confirms the stationarity of residuals and presents further evidence towards homoskedasticity.

## 4.5 Gauss–Markov Conditions

Although we present strong evidence towards all Gauss–Markov conditions being satisfied, we further confirm our findings using the tests outlined in Section 3.2.

- **Linearity:** The model is linear in parameters by design (Equation ?? & Equation 1).
- **Full Rank:** We computed a lack of perfect multicollinearity with  $VIF_{CPI,u} = 1.023 < 5$  and  $\text{cond}(X) = 14.75 < 30$
- **Spherical Errors:** Results of the BP test yielded  $BP = 3.88$  and  $p = 0.73$  failing to reject the null hypothesis of homoskedasticity. Additionally, BG tests for lags  $[1, 8]$  (2 years) all failed to reject the null hypothesis of no autocorrelation with the most significant result being  $BG(7) = 6.14$  and  $p = 0.48$ ; while the least significant was  $BG(3) = 5.19$  and  $p = 0.61$ .
- **Strict Exogeneity:** While strict exogeneity is not testable, the BG test and correlation analysis of  $X$  support the weak exogeneity assumption. Therefore, we elect to proceed with the assumption of exogeneity given the results of all prior diagnostics. The discussed final check of Harvey–Collier test results in  $HC = 0.83$  and  $p = 0.41$  failing to reject the null hypothesis of  $\mathbb{E}[w_i] = 0 \forall w$ .

## 5 Conclusion

With the help of normalization of variables, we were able to construct a well-behaved  $ARX(1, 1)$  model that, against expectations, showed no evidence against BLUE. Although the model showed signs of overfitting in-sample, the statistical tests and equality of variable distributions support the conclusion of an extremely interpretable estimator where coefficients may show connections to empirical elasticities of the used Keynesian regressors.

An unexpected finding was made with the unemployment coefficient, but the model as a whole shows enough stability and in-sample goodness-of-fit to warrant further comparison of empirical coefficients to economic theory. Moreover, we were able to display the use new and developing statistical tests and adapt to the unique challenges brought by the time-series context.

All in all, we believe that our model provides a decent starting point for further research and analysis. While the limited out-of-sample testing did not yield results that immediately display predictive instability, we strongly believe that as more data becomes available, the model will exhibit the well-known symptoms of overfitting. Therefore, we want to stress that the idea behind the creation of this model is to explore where theory and empirical evidence meet, and we encourage future researchers to build upon our findings.

## 6 Remarks

### 6.1 Remark 1: VIF Application on a Reduced Design Matrix

Although VIF just a measure and has no constraints on its application, the conventional use is to check for multicollinearity among the complete  $X$  matrix. In models with AR components, this application of VIF leads to two issues (assuming the dependent variable is actually autocorrelated):

- The AR components inside  $X$  will create processes equivalent to the an AR estimator of  $y$ . To demonstrate, let's assume a relationship  $y_t \sim y_{t-1}$  exists in an  $AR(2)$  model. As shifting the variable in time does not change the assumed relationship, we also have  $y_{t-1} \sim y_{t-2}$ . As a consequence, the design matrix (which is composed of  $y_{t-1}$  and  $y_{t-2}$ ) will exactly replicate an  $AR(1)$  estimator of  $y$ . With lags selected to maximize  $R_y^2$ , we end up also maximizing the  $R^2$  in the VIF calculation.
- The point above extends to exogenous variables as well. Again, assuming  $y_t \sim x_{t-1}$  exists, we also have  $y_{t-1} \sim x_{t-2}$ . Although  $y_{t-1} \sim x_{t-2}$  never directly occurs, the design matrix in this case produces  $y_{t-1} \sim x_{t-1}$ . This is not an exact replication of the designed relationship, but by autocorrelation of  $y$ , it is a very safe assumption that  $y_{t-1} \sim x_{t-1}$  will produce an  $R^2$  comparable to the original estimator. As a result, we will again be maximizing the  $R^2$  in the VIF calculation if we're constructing the model to explain the dependent variable as best as it can.

The points discussed show the well-known fact that VIF is not designed for these scenarios. Therefore, many studies, alongside this paper, elect to use VIF on a reduced  $X$  matrix stripped of AR components. (Niu and Li, 2022 & Karlström et al., 2023)

### 6.2 Remark 2: Time-Series Application of KS Test

The i.i.d. assumption of the classic KS test was addressed by Præstgaard with a bootstrapping approach. Yet, the examples, lemmas, and theorems presented in the paper were derived from row-exchangeable samples.

With row-exchangeability, Præstgaard was able to apply standard bootstrap resampling while preserving the continuity of the empirical distribution. However, time-series data inherently requires the row-structure to be preserved; hence rendering the standard bootstrap inappropriate. To preserve the time-dependent dynamics, our methodology elected to use a **Block Bootstrap** approach. This specific adaptation is not documented nor theoretically justified in Præstgaard's work; consequently, while we regard it as a reasonable and practically reliable extension for the purposes of this study, it remains an empirical choice that invites further theoretical investigation.

A block bootstrap 1-sample KS test was derived in a recent preprint paper (Chandy et al., 2025) and applications of bootstrap techniques in KS style tests were demonstrated in multiple studies. (Psaradakis, 2003 & Wyłupek, 2023) These works place bootstrap KS tests on a more solid theoretical footing, even though our specific use case is undocumented.

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